MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, EXTRA FLEXIBLE, CONVOLUTED, FLUORINATED ETHYLENE PROPYLENE, CLOSE CONVOLUTION

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the tubing described herein shall consist of this document and the issue in effect of MIL-T-81914 (AS).

REQUIREMENTS:

Convolution type: Helical (see 3.3)

Class:

Class 1 - Standard wall thickness
Class 2 - Thin wall thickness

Construction details: Construction details, including available sizes, dimensions and tolerances, are located in Table I. Lengths shall be as specified by the procuring activity.

Continuous operating temperature: -67°C (-88°F) to +200°C (392°F)

Color: Unless otherwise specified, the supplied color shall be clear, natural.

Physical properties: General physical requirement values along with associated test conditions are located in Table II.

Military Part Number: Consists of the basic number of this specification sheet and a dash number as shown below:

M81914/4-0000

- Identifier (Table I)
- Class when applicable (when no classes are specified, the numeric 0 shall be used)
- Type of convolution (1- helical, 2- annular)
- Specification sheet

Close convolution, Fluorinated ethylene propylene, thin wall, 0.500 inch is identified as M81914/4-1206.

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TABLE I

CONSTRUCTION DETAILS

Identifier	Maximum Inside Dia.	Minimum Inside Dia.	Maximum Outer Dia.	Wall Thickness Max.		Convolutions per inch; ± 1		Weight per 100 ft., Max.		Min. Bend Radius
				Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	
**01	.188	.181	.320	.018	.013	10	11	1.7	1.4	0.31
**02	.281	.273	.414	.018	.013	10	11	2.0	1.6	0.41
**03	.312	.306	.450	.018	.013	10	11	2.1	1.9	0.41
**04	.375	.359	.510	.018	.013	10	11	2.5	2.0	0.50
**05	.437	.427	.571	.018	.013	10	11	3.9	2.9	0.50
**06	.500	.480	.650	.023	.018	9	10	4.6	3.4	0.75
**07	.625	.603	.770	.023	.018	9	10	5.5	4.4	0.75
**08	.750	.725	.930	.023	.018	8	9	6.9	5.4	0.93
**09	.875	.860	1.073	.023	.018	7	8	8.9	6.9	1.25
**10	1.000	.970	1.226	.023	.018	7	8	9.5	7.5	1.25
**11	1.125	1.105	1.390	.023	.018	7	8	10.5	8.5	1.43
**12	1.250	1.205	1.539	.023	.018	6.5	7	11.2	9.7	1.43
**13	1.500	1.437	1.832	.023	.018	5.5	6	12.0	11.3	1.75
**14	1.750	1.688	2.082	.023	.018	5.5	6	13.5	12.3	2.00
**15	2.000	1.937	2.332	.023	.018	5.5	6	15.5	13.3	2.25
**16	6.000	5.870	6.120	-	.018	-	6	-	32.1	11.0

** The asterisks shall be replaced by convolution type and class designation

NOTE: Unless otherwise specified, all dimensions are in inches.

TABLE II

PHYSICAL REQUIREMENTS

PROPERTY	REQUIREMENT	TEST METHOD
Construction details, inches	In accordance with Table I	4.6.1
Stress in psi at 10% strain	Class 1- 250 to 270 Class 2- 200 to 500	4.6.2
Specific gravity, max.	2.18	4.6.3
Crush resistance, lbs., Horizontal, min.	Class 1-10; Class 2-7	4.6.4

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TABLE II (Continued)

PROPERTY	REQUIREMENT	TEST METHOD
Low temperature flexi-	No cracking	4.6.5; 10,000 cycles
Heat shock	No dripping, flowing or cracking	4.6.6; 175 ±1°C (347 ±2°F) for 4 hours
Longitudinal change %, max.	15	
Heat aging		4.6.7; 175 ±1°C (347 ±2°F) for 96 hours
Stress in psi @ 10% strain, % min. of init. value	80	
Fluid resistance		4.6.8
Stress in psi @ 10% strain, min. % of init. value	80	
Flammability	Not flammable 1/	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	10,000 volts no breakdown.	4.6.11

1/ See 6.4 "Properties not tested"

Standard Procurement: Procured as coils. Unless otherwise specified, individual lengths shall be 18 feet.

Preparing Activity
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